Scheduling Criteria and Metrics

The CPU scheduler selects a process from the Ready queue and allocates CPU cycles. Performance metrics include:

1. CPU Utilization: Percentage of time CPU is executing user or system processes.
2. Throughput: Number of processes completed per unit time.
3. Turnaround Time: Total elapsed time from submission to process completion.
4. Waiting Time: Aggregate time spent waiting in the Ready queue.
5. Response Time: Time from request submission to first output response.

First-Come First-Served (FCFS) and Shortest Job First (SJF)

FCFS is non-preemptive: processes are assigned CPU in arrival order. A long CPU-bound process arriving before short I/O-bound processes leads to the Convoy Effect, drastically inflating average waiting time.

SJF associates each process with the length of its next CPU burst. Provably optimal in minimizing average waiting time. Preemptive SJF is known as Shortest Remaining Time First (SRTF).

Round Robin (RR) and Priority Inversion

Round Robin assigns each process a fixed time quantum q. If a process does not complete within q, it is preempted and returned to the tail of the Ready queue. If q is excessively large, RR degenerates into FCFS; if q is too small, context switch overhead dominates CPU execution.

Priority Inversion occurs when a lower-priority process holds a mutual exclusion lock required by a higher-priority process, while an intermediate-priority process preempts the lower-priority one. Solved using Priority Inheritance Protocol.